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AI-powered ensemble machine learning to optimize cost strategies in logistics business

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ABSTRACT

This research investigates the potential advantages of using artificial intelligence (AI) to drive ensemble machine learning (ML) for enhancing cost strategies and maximizing profits. This study aims to explore the ability of AI-powered ensemble ML to optimize cost strategies by simulating business threshold cost data to determine optimal mitigation strategies. The dataset comprises 6561 potential tuples, and three ensemble ML methods are employed as ML algorithms to identify patterns and relationships in the cost data for strategic decisions. The originality of this project lies in its demonstration of the capacity of simulated data to enhance cost-saving strategies for businesses. This research contributes to the existing literature on AI and ML applications in business by revealing the potential of ML applications for business owners and personnel involved in production and marketing. The findings of this research have significant implications for a wide range of industries, including transportation, logistics, and retail.

1. Introduction

The global logistics delivery market is expected to grow at a compound annual growth rate of 8.5% from 2022 to 2027 (Melton, 2022). Increasing e-commerce sales drive the market, the growing demand for express delivery services, and the rising adoption of contactless delivery (Al-Adwan & Yaseen, 2023; Chen & Yang, 2022; Jiang et al., 2023). The growth of e-commerce is one of the key drivers of the logistics delivery market (Morganti et al., 2014). As more and more people shop online, the demand for intelligent logistics delivery services is increasing (Spadafora et al., 2022). The growing demand for express delivery services is another key trend in the logistics delivery market (Martin et al., 2021). Customers increasingly demand faster delivery times, which drives the demand for express delivery services (Pereira Marcilio Nogueira et al., 2022). Express delivery services typically offer guaranteed delivery times, which is essential for businesses that need to get their products to customers quickly.

The rising adoption of contactless delivery is a new trend that has emerged in the wake of the COVID-19 pandemic (Wang et al., 2021). The pandemic has increased concerns about the spread of germs, leading to the adoption of contactless delivery services (Milewski & Milewska, 2021). Contactless delivery services allow customers to receive their packages without having to come into contact with delivery personnel. The global logistics delivery market is expected to reach $622.69 billion in revenue by 2029 (Research, 2023). This growth is driven by the abovementioned factors and the increasing demand for logistics delivery services from businesses and consumers (Fehn et al., 2023). The logistics delivery market is highly competitive, and the key players are constantly innovating to stay ahead of the competition (Kulkarni et al., 2022).

However, this growth also pressures logistics carriers to optimize their cost structures to remain competitive. Traditional cost optimization methods, such as manual route planning and static pricing models, are needed to meet the demands of modern business (Schmitt, 2023). These methods are often slow, inflexible, and unable to adapt to the industry's dynamic nature. There is a need for new and innovative approaches to cost optimization in the logistics business. Artificial intelligence (AI) ensemble machine learning (ML) can revolutionize the industry by providing a more efficient, flexible, and adaptable way to optimize cost strategies (Verma et al., 2021). The current cost optimization methods used by logistics carriers are inefficient, inflexible, and unable to adapt to the industry's dynamic nature. This is leading to increased costs and decreased profitability.

The logistics business is a data-intensive industry with vast information, including route details, customer demand patterns, and fuel

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consumption metrics (Khan Md. Hasib et al., 2023). Traditional cost optimization methods often need to capture this data's complex relationships and hidden patterns (Sundari & Subaji, 2020). To address this challenge, AI-powered ensemble ML can effectively analyze this data and identify these hidden patterns, providing valuable insights for optimizing cost strategies. This methodology can adapt to dynamic market conditions, which often include unpredictable changes like fluctuations in demand, fuel prices, and weather patterns (Schmitt, 2023). AI-powered ensemble ML can continuously monitor these changes and adjust cost strategies in real time, ensuring that logistics carriers remain competitive and profitable in a dynamic environment (Wang et al., 2023). In addition, accurate demand forecasting is crucial for efficient resource allocation and cost optimization in logistics.

The integration of AI and ML in the logistics industry has the potential to bring about transformative changes in our daily lives. The impact of these cutting-edge technologies will be tangible, making faster deliveries, lowering shipping costs, and providing a more comprehensive selection of products at competitive prices (Chen & Yang, 2022; Wang et al., 2023). The use of AI and ML in logistics also promises to enhance the efficiency of businesses, reduce their costs, and improve customer satisfaction, ultimately leading to a more profitable venture (Yaiprasert & Hidayanto, 2023). The benefits of AI and ML are not limited to consumers and businesses alone. Integrating these technologies into the logistics industry will also profoundly impact the economy (Bughin et al., 2018). It will create new job opportunities that require specialized skills and propel the growth of other industries that support the logistics sector. Additionally, deploying AI and ML in logistics will help reduce the carbon footprint associated with transportation, making it an eco-friendly option.

Despite their simplicity in implementation, single-method ML approaches often need help with multiple drawbacks that limit their effectiveness and generalizability (Verma et al., 2021). These limitations include overfitting, underfitting, and hyperparameter sensitivity (Beltrán-Royo et al., 2023). Overfitting occurs when a single ML model learns the idiosyncrasies of the training data too closely, leading to poor performance on unseen data (Aceña et al., 2022). Conversely, underfitting occurs when a single ML model fails to capture the complexities of the data, resulting in insufficient predictive power. The performance of single ML models is often susceptible to the selection of hyperparameters, which can significantly degrade the model's performance. To address these limitations, ensemble ML methodologies that utilize multiple ML models offer a promising approach.

AI-powered ensemble ML can accurately analyze historical data, current trends, and external factors to predict demand fluctuations (Khan Md. Hasib et al., 2023; Rosário & Dias, 2023). This enables logistics carriers to proactively adjust their operations, such as staffing, fleet management, and warehouse optimization, to meet demand efficiently (Dumas et al., 2018). Furthermore, route planning and resource allocation are critical components of cost optimization in the logistics business. AI-powered ensemble ML can analyze road networks, traffic patterns, and customer locations to generate optimized routes that minimize fuel consumption, delivery times, and overall costs (Schmitt, 2023). Additionally, AI algorithms can optimize resource allocation across the logistics delivery network, ensuring that vehicles, personnel, and facilities are utilized efficiently (Kumar et al., 2021). Finally, AI-powered ensemble ML provides logistics carriers a powerful tool for making informed, data-driven decisions. By analyzing vast amounts of data and identifying hidden patterns, AI algorithms can provide actionable insights that improve decision-making across all logistics delivery operations.

The logistics delivery business is a complex and challenging industry. Firms in this industry face several issues and problems, including lost or damaged logistics, late deliveries, high costs, competition, regulations, and technology (Ouyang et al., 2022; Zhang et al., 2022). These challenges can make it difficult for businesses to operate and stay competitive. However, by understanding these challenges, companies can make informed decisions about managing their businesses and succeeding (Leyerer et al., 2019; Tavakoli et al., 2022).

Business costs are essential to track because they help companies determine their profitability. By understanding their costs, businesses can decide how to price their products and services, allocate resources, and improve their profitability (Bilal & Oyedele, 2020; Ding et al., 2022; Dumas et al., 2018). Therefore, tracking technology is an essential tool for businesses that are looking to control their costs and improve their bottom line (Wang et al., 2023; Zhao et al., 2020). Tracking business spending is essential for several reasons. It can help businesses see where their money is going and identify areas where they can cut back.

AI can close these problem gaps in business spending in several ways (Dondapati et al., 2021; Yaiprasert & Hidayanto, 2023). AI can automate tracking spending, freeing time to focus on other tasks. This can help ensure that spending is tracked accurately and efficiently. AI can identify anomalies and patterns in spending, which can help identify areas of overspending or potential problems (Rosário & Dias, 2023). AI can recommend improving expenditures, such as reducing unnecessary expenses or reallocating spending to more productive areas (Sharma et al., 2021). Some specific examples of how AI can close problem gaps in business spending. A company that sells online could track its spending on marketing campaigns (Cuzzocrea et al., 2020). AI could be used to identify which campaigns are the most effective and recommend how to improve the performance of less effective campaigns. A manufacturing company could track its spending on raw materials (Zheng et al., 2022). AI could be used to identify trends in the cost of raw materials and make recommendations about negotiating better prices with suppliers. AI can automate inventory management and quality control (Praveen et al., 2019). A retail company could track its spending on inventory (Zhang et al., 2018). AI could be used to identify slow-moving inventory and to make recommendations about how to reduce the amount of stock that is sitting on the shelves (Wang et al., 2023; Xiong et al., 2021).

This study was motivated by the need for adaptability and resilience to compete in a highly competitive market. It provides an alternative approach to cost strategy in the logistics delivery industry. The logistics delivery industry is highly competitive, and businesses constantly look for ways to reduce costs and improve profitability. Traditional cost strategies, such as manual labor and spreadsheets, are often inefficient and time-consuming, leading to missed opportunities and lost profits. Collaboration AI offers a promising new approach to cost strategy for businesses. Ensemble AI can automate tasks, analyze data, and make predictions (Khanday et al., 2022). This can help businesses to close the gap on traditional cost strategies and achieve significant cost savings.

This study identifies research gaps in the use of ML and simulated data in business contexts, with a specific focus on the underutilization of ensemble ML to optimize cost strategies. The research investigates the potential impact of this approach on a logistics delivery company's business operation, highlighting potential risks and challenges. It provides valuable insights and tactics to effectively address the challenge and develop a thorough understanding of it. The research analyzes the impact of different strategies on resolving the challenge, guiding decision-making. The expected outcomes of this research project are to identify how AI technology can increase revenue and provide recommendations on how companies can use AI technology to remain competitive in an increasingly crowded market. The study aims to provide readers with a better understanding of how AI and ML can be combined to create more robust digital marketing tools and maximize revenue.

On the other hand, this paper focuses on demonstrating the effectiveness of AI-powered ensemble ML in reducing costs and improving profitability within the logistics business. A cornerstone contribution of this study lies in developing a novel approach that harnesses ensemble ML techniques to optimize cost strategies in the logistics business domain. This approach seamlessly integrates multiple ML models, empowering the system to learn from diverse data sources and make more accurate predictions. Ensemble ML has been rigorously

demonstrated to enhance prediction accuracy and mitigate the risk of errors in the decision-making process. This research endeavor is expected to provide valuable insights into the efficacy of AI-powered ensemble ML as a tool for optimizing cost strategies in the logistics business. The present study aims to make significant contributions to the research community in the following ways:

(i) The study meticulously explores the optimization of cost strategies in logistics business scenarios by employing AI-powered ensemble ML.
(ii) It endeavors to develop a dynamic tracking model and decision-making mechanism for logistics delivery operations.

This study leverages AI to simulate business threshold cost data to achieve the objectives above and identify optimal mitigation strategies. The ML algorithms employed in this study, including decision trees, logistic regression, and nearest neighbor methods, effectively extract patterns and relationships within the cost data, enabling informed strategic decision-making. The remainder of this paper is structured as follows: Section 2 presents the hypothesis development process. Section 3 details the methodology for the ML algorithm and programming development. The results and analysis of this study are presented in Section 4. Section 5 discusses the study's contribution to the existing literature and its practical implications, and Section 6 provides the study's conclusion.

2. Hypothesis development

Technological advances have significantly changed the logistics delivery industry in recent years. The increasing demand for e-commerce has driven tremendous growth in the industry, prompting companies to seek ways to optimize their operations and reduce costs as competition grows. One emerging trend is using AI-powered ensemble ML algorithms to optimize cost strategies. This technology is making waves in the industry as it can increase logistics business profits. Therefore, this paper aims to develop hypotheses related to AI-powered ensemble ML to optimize cost strategies in the logistics business case. The study's primary objective is to explore the potential benefits of using AI-powered ensemble ML algorithms to optimize cost strategies in the logistics delivery industry. The study seeks to contribute to the existing literature by developing hypotheses to guide future research.

H1: An accurate and efficient cost-reduction strategy creates more significant profits.

Profit maximization is crucial for long-term success in the competitive logistics business industry. This study proposes that implementing an accurate and efficient cost-reduction strategy can significantly enhance profit margins for logistics businesses. The underlying assumption is that companies can boost their profitability by minimizing unnecessary expenses (Reyes & Hargreaves, 2022). AI-powered ensemble ML offers a powerful tool to identify areas where costs can be reduced. By analyzing vast amounts of data related to transportation, labor, and packaging costs, AI algorithms can pinpoint opportunities for cost optimization (Cesar Perez Lopez, 2021). By implementing effective cost-reduction strategies informed by AI-powered insights, logistics businesses can streamline operations, enhance efficiency, and achieve greater profitability. The central hypothesis of this research is that an accurate and efficient cost-reduction strategy, driven by AI-powered analytics, leads to more significant profits for logistics businesses. Companies that can optimize delivery routes, minimize fuel consumption, and reduce waste through AI-powered insights will gain a competitive edge in the market.

H2: AI is an effective tool that can be used to manage costs.

In the dynamic and cost-sensitive logistics industry, efficient cost management is essential for maintaining profitability and competitive advantage. This study proposes that AI-powered ensemble ML is a powerful tool for effective cost management. The underlying assumption is that AI algorithms can analyze vast amounts of data to uncover hidden patterns and trends that may remain undetected by traditional methods (Sundari & Subaji, 2020). By leveraging AI-powered ensemble machine learning, logistics businesses can gain valuable insights into their operations and decision-making processes, enabling them to identify and address areas of cost inefficiency (Rawat et al., 2021). For instance, AI algorithms can pinpoint specific routes or operational procedures that lead to excessive fuel consumption or vehicle wear and tear, allowing businesses to optimize operations and minimize unnecessary expenses. Additionally, AI can reveal hidden costs often overlooked in traditional cost-reduction strategies, such as inefficient packaging or underutilized resources. By analyzing data from various sources, including sensors, delivery routes, and customer behavior, AI algorithms can identify these hidden costs and provide businesses with actionable insights for cost optimization. Consequently, AI-powered ensemble ML emerges as a transformative tool for effective cost management in the logistics business, enabling companies to achieve greater profitability and operational efficiency.

H3: AI collaboration to track production costs will produce more accurate results, which leads to intelligent strategy adjustments.

Cost management remains critical to maintaining competitiveness and profitability in the ever-evolving logistics industry (Wang et al., 2023). This study proposes that AI-powered ensemble machine learning, when harnessed collaboratively, can revolutionize cost tracking and strategy optimization. By collaborating with AI algorithms, the underlying assumption is that businesses can gain deeper insights into their production costs, enabling them to make intelligent and timely adjustments to their cost management strategies. For instance, AI algorithms can continuously analyze production data to identify where costs escalate or deviate from expected patterns (Singh et al., 2022). This real-time information empowers businesses to make strategic adjustments to their production processes, such as optimizing resource allocation, minimizing waste, and negotiating better supplier rates, ultimately reducing costs. Furthermore, AI-powered ensemble ML can equip companies with the agility to adapt to dynamic market conditions (Rosário & Dias, 2023). By analyzing customer demand, weather patterns, fuel prices, and other relevant factors, these algorithms can provide real-time insights that enable companies to proactively adjust their operations, pricing models, and delivery routes. This real-time adaptation to market fluctuations can significantly enhance a company's competitive edge in a rapidly changing environment.

3. Methodology

The requirements of logistics delivery companies determined the study design. Data on production and customer costs were considered to create a model, assigning different outcomes for decision-making. An association rule was used to determine key and value data and then submitted to the ML training process. Verification of the system was designed to be at least 85% accurate as a standard while adding one more ML method for iteration further increased the confidence in the usability of the decision-making system. The research framework is described in Fig. 1.

Generating simulated data is a crucial step in the AI workflow loop, which allows researchers to create ground truth for measuring model accuracy by designing computer-generated datasets customized to the AI system's specific application. This ensures the AI system learns the right features and makes the correct predictions. Simulated data is a valuable tool for AI development to improve AI system's accuracy, scalability, privacy, safety, and cost-effectiveness. Simulated data is often the only option for training AI systems, especially when real data is unavailable, too expensive, or time-consuming to collect. In other cases, simulated data can be used in conjunction with real data to improve the accuracy and performance of AI systems.

The association rule is a rule-based ML method for discovering interesting relationships between variables in large datasets. It identifies

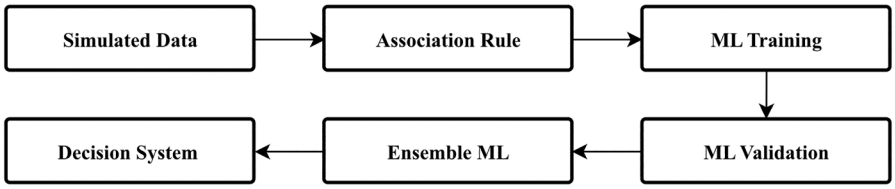

Fig. 1. Research model framework.

strong rules located in databases using some interestingness measures. The association rule is used to find the rules that determine how or why certain items are connected in any given transaction with a variety of items. After creating an association rule for a dataset, researchers need to use ML training to teach their AI models how to recognize these patterns. ML training involves feeding data into a model so that it can learn from examples provided by the user. The more data used to train the model, the more accurate it becomes. However, it is essential to use high-quality simulated data representative of the real-world problem the model is trying to solve.

ML validation is an essential step in the ML process that helps to ensure that the model is accurate and can be used to make reliable predictions. After training an ML model, it is essential to validate its accuracy using a separate dataset that the model has not seen before. The model's predictions on this new data can be compared to the actual outcomes to see how accurate the model is. Ensemble ML is a technique that combines the predictions of multiple ML models to produce a more precise prediction. It is based on the principle that a diverse group of models is more likely to be accurate than any individual model.

Finally, after an ML model has been trained, validated, and re-validated, it is ready to be used in a decision system that uses an ML model to make decisions. The decision system typically takes in data and then uses the ML model to make a prediction or decision.

2.1. Machine learning method

A combination of decision trees, logistic regression, and nearest neighbors can be used as an effective method for ensemble and firming the results of ML classification tasks. The decision tree is used first to build a classification model with the logistic regression, and the nearest neighbors model is then applied to repeat the classification results. By combining these three methods, the bias or errors in each model can be mitigated, making them more accurate and reliable. Additionally, some advanced ML techniques, such as ensemble methods (e.g., bagging and boosting), can also be used to combine the outputs to improve the overall result.

2.1.1. Decision tree

Decision trees are a powerful and popular technique used in AI to solve complex problems. They provide a way to determine the best action for a given situation systematically. Decision trees are structured like flowcharts, with branches representing possible decisions and outcomes. The decision tree searches for the most "correct" decision at each branch based on pre-defined criteria. Each branch has a weight associated with it, based on how likely a particular decision will lead to the desired outcome (Sheppard, 2017). For example, the decision tree would weigh each option's success probability if two possibilities exist – "Go left" or "Go right". Decision trees are constructed using a scoring function at the most basic level. This function takes inputs on the current state of the problem and assigns a score to each possible action. The action with the highest score is the one that should be taken. A typical scoring function $F(x)$ might take the following form $F(x) = w_1X_1 + w_2X_2 + \ldots + w_nX_n$, where $X_1, X_2, \ldots, X_n$ are variables describing the current state of the problem, and $w_1, w_2, \ldots, w_n$ are weights assigned to the variables that indicate their relative importance.

2.1.2. Logistic regression

Logistic regression is an ML algorithm used to address classification problems such as predicting whether or not a customer will buy a product or whether it is worth the investment. It is a statistical technique that models the relationship between a set of independent variables and a dependent variable. The dependent variable is usually a binary outcome, such as "yes" or "no" whereby the outcome can be modeled as a probability. Logistic regression estimates the dependent variable's probability given the independent variable's values.

Logistic regression uses an equation to estimate probabilities known as the logit function. It takes the form $p = 1/(1 + e^{-z})$, where p is the probability of the dependent variable, e is the base of natural logarithms (approximately 2.718), and z is the difference between the linear predictor function and threshold value. The linear predictor function is a combination of the independent variables multiplied by their coefficients, $\theta_0 + \theta_1X_1 + \theta_2X_2 + \ldots + \theta_nX_n$. The output of the logit function ranges from 0 to 1, with higher probabilities indicating a better fit with the data. The coefficients must be estimated to predict the dependent variable's probability. This is typically done using maximum likelihood estimation, which searches for the coefficient values that maximize the log-likelihood of the data.

2.1.3. Nearest neighbors

The nearest neighbor is an ML method that classifies data points or objects. It compares a given data point/object to its nearest neighbors and assigns it to the category most of its neighbors belong to. The principle behind the nearest neighbor algorithm is that similar objects are close together. This means that the nearest neighbors of a particular object can be used to identify its class or label. The nearest neighbor is a simple and intuitive algorithm, but its effectiveness depends on the distance metric. Two standard distance metrics used in the nearest neighbor algorithm are Euclidean and Manhattan distance.

Euclidean Distance: the straight-line distance between two points in an Euclidean space. The formula for calculating the Euclidean distance between two data points $p1$ and $p2$ is given as $d(p_1, p_2) = \sqrt{[(p1_1 - p2_1)^2 + (p1_2 - p2_2)^2 + \ldots + (p1_n - p2_n)^2]}$, where $p1_1, p1_2, \ldots, p1_n$ are the coordinate values of point $p1$ and $p2_1, p2_2, \ldots, p2_n$ are the coordinate values of point $p2$.

Manhattan distance: the sum of absolute differences between the coordinates of two data points. The formula for calculating the Manhattan distance between two data points $p1$ and $p2$ is given as $d(p_1, p_2) = |p1_1 - p2_1| + |p1_2 - p2_2| + \ldots + |p1_n - p2_n|$, where $p1_1, p1_2, \ldots, p1_n$ are the coordinate values of point $p1$ and $p2_1, p2_2, \ldots, p2_n$ are the coordinate values of point $p2$.

2.2. Programming language

Mathematica is a powerful programming language used in AI and ML (Bernard, 2021). It provides an interface and functions for developing algorithms, writing code, and managing data. AI and ML implementation with Mathematica can analyze large datasets, understand complex correlations, and build accurate predictive models in minutes. Mathematica's AI/ML capabilities are powered by its advanced symbolic computation library, allowing users to manipulate symbolic expressions to solve problems. This study used Mathematica Trial Version 13.0.1 of

the Wolfram language, run on a MacBook Pro with an M1 processor and 8 GB of RAM.

2.3. Model

2.3.1. Trade-offs and biases

As part of this research, the use of simulated data was explored. Simulated data can provide a realistic learning environment for an AI system by presenting multiple datasets from different sources, which may include real-world environments or simulations. The simulated data can also help the AI system understand correlations and patterns among data sources. Additionally, the performance of the AI system can be evaluated using simulation data, which can further aid in its development and modifications.

Using simulated data in AI research can be highly advantageous, but knowing the potential trade-offs and biases arising from its use is crucial. One of the significant concerns in this regard is the data representation, as simulated data may only partially reflect the complexities of the real world. This, in turn, can lead to less generalizable models for real-world data. Furthermore, the simulated data quality is paramount, as inaccurate or non-representative data can result in biased models. Lastly, the performance of the models can be affected by the amount and quality of the training data. If the training data is limited or biased, it could lead to less robust and generalizable models. Therefore, it is imperative to carefully consider these factors while using simulated data in AI research.

2.3.2. Prepare the data

In this study, data was generated to simulate the demands of a company based on its primary production cost history data. This allowed the creation of datasets that closely resemble real-world data. ML can also use simulated data to build realistic models and make predictions. The simulation will enable researchers to study the behavior of a system by creating a model consisting of a structure and rules governing this structure to produce various outputs. The technique presented in this study is used to explore the potential application of the business threshold value of cost (BTVC) to determine a simulated dataset for machine learning. This knowledge can assist in optimizing the accuracy of ML models and reduce the associated costs.

This research seeks to assess the feasibility of using the BTVC to evaluate or simulate data for ML projects and analyze the results. Specifically, it explores how the selected variables, such as cost possibility, affect the results and inform the trade-off between cost and risk. Finally, it explores the potential applications of the BTVC-based simulation of various possible cost tuples. An initial statement of the BTVC is shown in Table 1.

The BTVC consists of approximately 6,561 possible tuples, such as {-1, -1, -1, -1, -1, -1, -1, -1}, {0, 0, 0, 0, 0, 0, 0, 0}, and {1, 1, 1, 1, 1, 1, 1, 1}, etc., with rating scales ranging from -8 to 8, such as extremely risky, high risk, risk, poor, somewhat poor, fair, moderate, passable, neutral, acceptable, satisfactory, good, very good, excellent, outstanding, exceptional, and superlative, respectively. This study mixed the number of possible BTVC tuples on the decimal scale to provide sufficient learning data for ML, resulting in increased model accuracy, such as {1, 0.01, 1, 0.02, 1, 0.03, 1, 0.04}, {-1.0, -0.8, -0.6, -0.4, 0.4, 0.6, 0.8, 1}, etc. The total possible tuples of the BTVC are used for 14,211 cases. This means that the threshold value can be set at the decimal level. This process demonstrates that the company can set the BTVC value in decimals from -1 to 1 or apply any range of threshold values as required. Furthermore, the company can determine each cost type in other amounts as necessary.

2.3.3. Divide the data into training sets

Once the data has been prepared, it must be divided into training and test sets. The training dataset will be used to develop the model, and the test dataset will be used to measure its performance (Aggarwal, 2016). An association rule learning algorithm, a supervised ML technique, is employed for the training dataset. This algorithm is used to identify relationships between items in large datasets, often for cost analysis. It is characterized by two measures: "support" and "confidence" to ascertain the frequency of investment items and the likelihood of them being purchased together, respectively. The goal of association rule learning is to identify and generate rules that explain correlations and patterns in data. An example algorithm for generating training data with association rules in ML processes is presented in Code Snippet 1. The process aims to define the relationships between different datasets or variables by mining databases for patterns, which can then be used to create practical algorithms or models for forecasting outcomes from any given set of variables.

```
Input1:  cases = Tuples[{-1, 0, 1}, 8];
Input2:  trainSet = Table[Which[ Total[cases[[i]]] == -8, Total[cases[[i]]] -> "01-
         Extremely risky", Total[cases[[i]]] == 8, Total[cases[[i]]] -> "17-
         Superlative"], {i, 1, Length[cases]}];
```

Code Snippet 1 "trainSet" is a sample training data. The association rule shows a correlation between the BTVC and the rating considered.

2.3.4. Choose and train an ML model

Several techniques exist for fitting an ML model to a given dataset and optimizing it through its loss function. This study uses decision trees, logistic regression, and nearest neighbor ML algorithms. The data should then be pre-processed into training, validation, and test datasets. Finally, the ML model should be trained by feeding the training dataset into the model and supplying it with the correct labels for each example to predict labels for unseen data accurately. This typically involves evaluating several parameters (Josephine Isabella & Srinivasan, 2018).

The selection and training of an appropriate ML model are essential to achieve maximum predictive power while avoiding overfitting. Perfect accuracy cannot be expected from the start, so it is necessary to test and validate the algorithm after each training cycle to verify its accuracy. ML training is an iterative process and must go through multiple cycles of training and validation before its performance can be deemed effective. Features and hyperparameters of the model can be modified to improve performance, for example, by adjusting the complexity of the model while testing different algorithms to increase accuracy. In this study, the training method used is outlined in Code Snippet 2.

```
Input1:  decisionTree = Classify[trainSet, Method ->"DecisionTree"];
Input2:  LogisticRegression = Classify[trainset, Method->"LogisticRegression"];
Input3:  nearestNeighbors = Classify[trainSet, Method ->"NearestNeighbors"];
```

Code Snippet 2 "trainSet" is a dataset for training each ML method.

2.3.5. Test the model

After training an ML model, it is essential to evaluate its accuracy. The most effective approach to achieve this is by testing the model's

Table 1
An initial statement of the BTVC.

Raw materials	Labor	Energy	Technology	Packaging	Distribution	Marketing	Premium product	Rating
-1	-1	-1	-1	-1	-1	-1	-1	Extremely risk
0	0	0	0	0	0	0	0	Neutral
1	1	1	1	1	1	1	1	Superlative

performance on unseen data, which provides a more accurate evaluation. This study uses a dataset created from random nesting to test the ML model. Random nesting is a widely used concept in AI, primarily in neural networks. It involves randomly organizing the structure of a system or algorithm, allowing it to handle tasks that require a higher degree of complexity. This is done by grouping data points and embedding them in a more extensive system, which can be easily manipulated to produce the desired output. One of the significant benefits of using random nesting for AI applications is that it reduces the amount of testing data required by the system. It enables the AI to choose a subset of data from a larger dataset and construct a smaller yet equally effective testing set. Code Snippet 3 shows a randomly generated dataset for testing the ML model, with the BTVC randomly ranging between -1.00 and 1.00 and a total of 1,000 BTVC.

```
Input1:  n = 1000;
Input2:  testSet = Table[If[i>n, Nothing, Table[If[i>8, Nothing, RandomChoice
         [{RandomInteger[{-1, 1}], RandomReal[{-1, 1}]}]], {i, 1, 8}]], {i, 1,
         1000}];
```

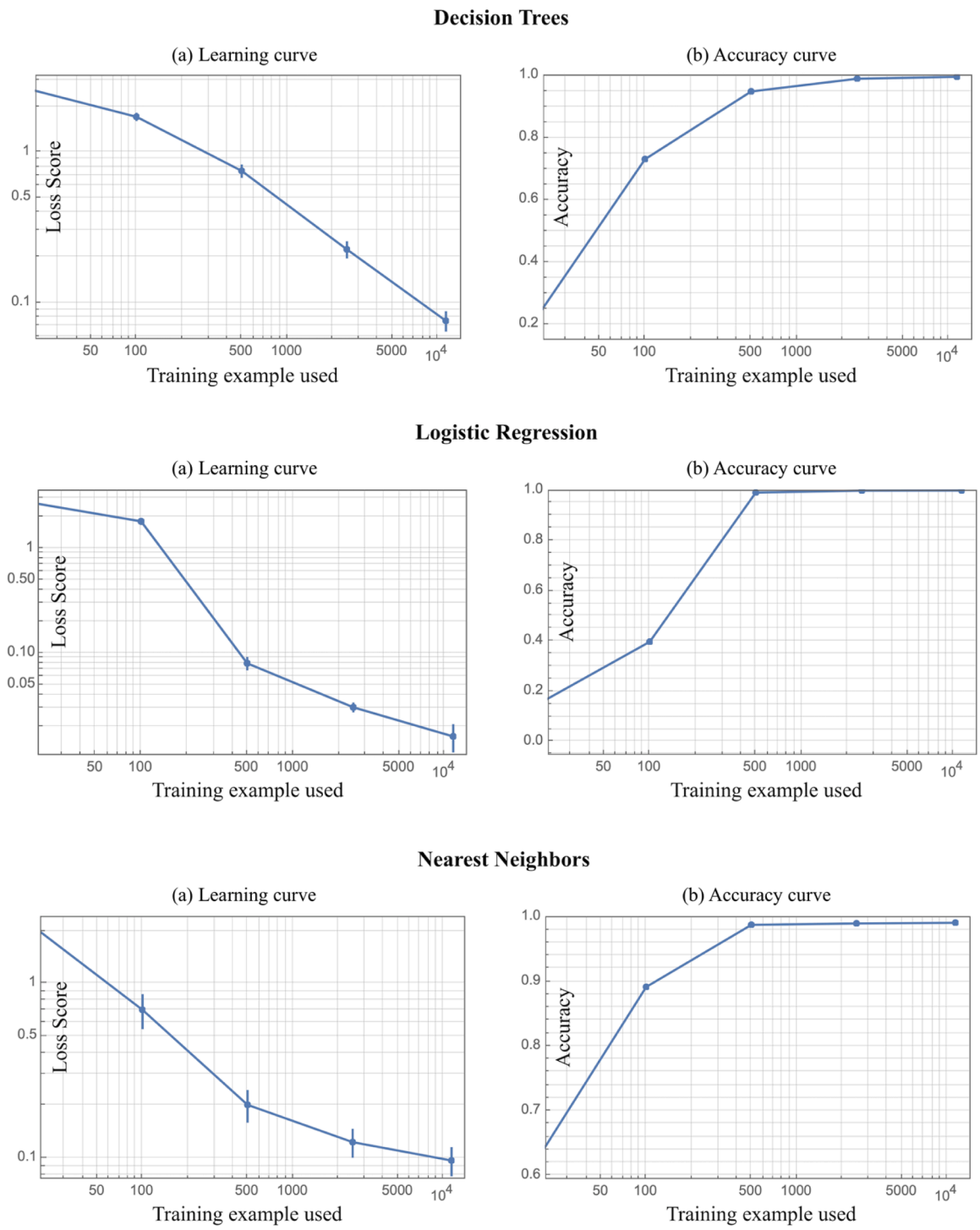

Fig. 2. Learned curve and accuracy curve of each ML method: (a) the learning curve has the x-axis representing the number of samples used and the y-axis representing the loss score, and (b) the accuracy curve has the x-axis representing the number of samples used and the y-axis representing the accuracy score.

Code Snippet 3 The BTVC value is a randomly generated nest for model testing.

2.3.6. Approach to validity and reliability

To ensure the validity and reliability of the research conducted, a series of steps were taken to address these challenges. The first step involved collecting simulated data from a comprehensive understanding of the logistics business and its associated data sources. This involved carefully crafting data to represent real-world scenarios and capture the complexity of the logistics delivery process. The next step was to extensively validate the simulated data to ensure its accuracy and consistency. This process included checks for data quality, completeness, and representativeness. The AI model's performance was then evaluated using various metrics, including accuracy, precision, recall, and F1-score. The models were also evaluated on unseen data to assess their generalizability. Finally, to ensure the finding's relevance and practical applicability, experts in the logistics business domain reviewed the study's conclusions. This multi-faceted approach to addressing validity and reliability challenges ensures that the research is high quality and provides practical insights for the logistics business.

2.3.7. Ensemble ML

The use of AI in various areas has become increasingly popular in recent years due to its potential for dramatically improving efficiency (Yaiprasert, 2021; Yaiprasert & Yusakul, 2022). One of the critical aspects of AI is that it can ensemble the results obtained from applying ML algorithms. As a result, AI is frequently used as a second and third layer of checks on the results obtained from ML techniques to ensure accuracy and reliability. By running an AI algorithm after the ML system has produced an answer, the accuracy of the response can be improved significantly. By repeating a different AI algorithm to validate the responses from the first one, the chances of detecting any mistakes or inaccuracies can be further increased.

4. Results

4.1. ML learning

The learning curve in ML provides a graphical representation of an AI algorithm's ability to learn from data over time. It plots the accuracy or loss score change against the number of iterations in a training dataset. The accuracy measures how well the algorithm predicted outcomes, while the loss score indicates how well it fits the data. An accurate ML curve is an inverse curve of the loss score over the number of iterations. The accuracy and loss scores decrease as the algorithm learns more from the data. Fig. 2 demonstrates the learning curve and the learning accuracy curve. Both graphs suggest that early learning incurs high loss scores and low accuracy. However, learning with around 500 data items yielded a loss score of less than 1, with an accuracy approaching 90%. The loss scores decreased to less than 0.1, indicating successful model training and excellent results.

The ML training results revealed that the decision trees method had a high accuracy of 99.46% ± 0.20% and a low loss score of 0.0755 ± 0.0100. The logistic regression method had the highest accuracy of 99.90% ± 0.08%, with the lowest loss score being 0.0159 ± 0.0014. The nearest neighbors method was also highly accurate, achieving an accuracy of 99.07% ± 0.34% with a low loss score of 0.0966 ± 0.0170, indicating a successful model application and excellent results.

The calibration curve represents a model's predicted probabilitie's reliability, enabling the comparison of observed data points with their predicted probabilities. The confusion matrix provides a tabular overview of the classification algorithm's performance, allowing users to evaluate the precision and recall of their model and identify which classes were misclassified.

Figs. 3 and 4 show the calibration curve and confusion matrix fitted with a perfect diagonal line, demonstrating that the model performs and provides excellent accuracy. The ML accuracy matrix plot results revealed that the decision trees, logistic regression, and nearest neighbor algorithms achieved an overall accuracy matrix of 100% with corresponding mean cross entropies of 0.00143, 0.00103, and 0.00979, respectively. These results demonstrate that the model could accurately identify the correct labels across all classes.

4.2. Model validation

Model validity represents the extent to which the computed behavior approximates the solution. Model validation evaluates an ML model's performance on new data, determining the model's precision, sensitivity, specificity, accuracy, and other metrics. This research uses cross-validation, a technique that involves dividing the original dataset into one or more smaller datasets for testing and training the ML model. Cross-validation is essential for evaluating predictive models as it helps prevent overfitting by avoiding train-test contamination. Accuracy, precision, recall, F-score, and AUC are commonly used metrics to evaluate ML models. The overall values of these metrics above 0.85 are crucial indicators of a model's performance. The confusion matrix in Fig. 5 and Table 2 demonstrate that the models in this study performed well and achieved excellence with quantitative performance measures above 0.95.

4.3. Ensemble ML

Ensemble ML is used to combine multiple models and improve the overall performance of a predictive model. The ensemble ML method used in this study combined three ML methods, resulting in validation accuracies ranging from 97 to 99%, thus demonstrating excellent

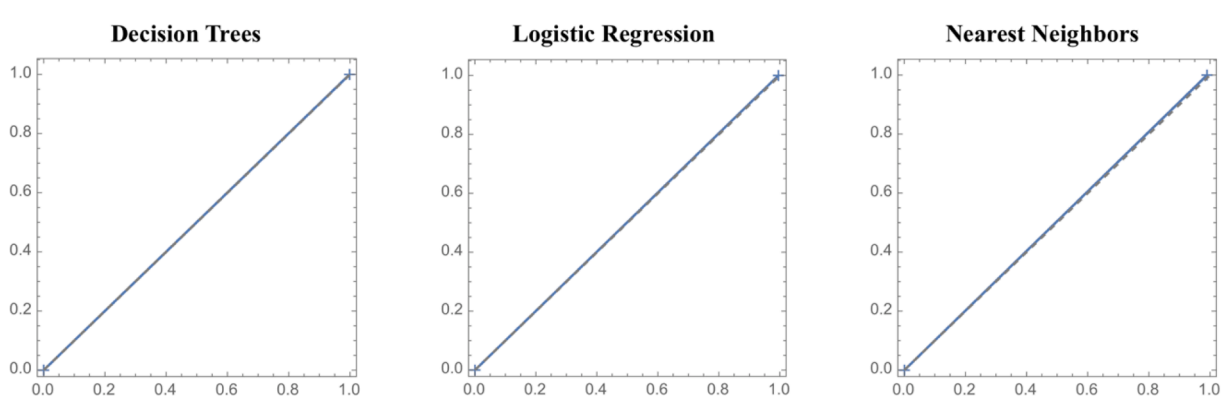

Fig. 3. The probability calibration curve of each ML training method has the x-axis representing the actual probabilities and the y-axis representing the predicted probabilities.

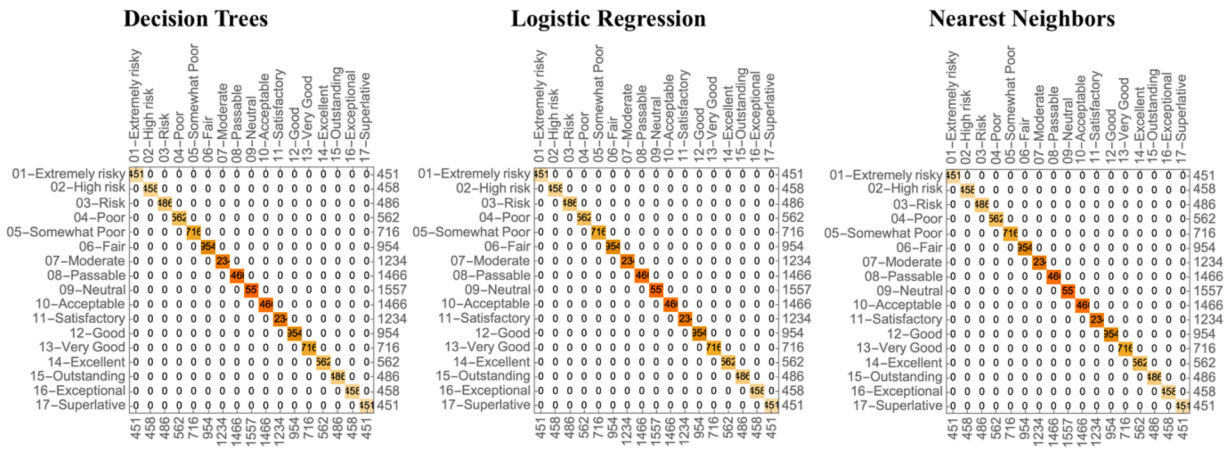

Fig. 4. The accuracy matrix plot of each ML training method has the x-axis representing the actual class and the y-axis representing the predicted class.

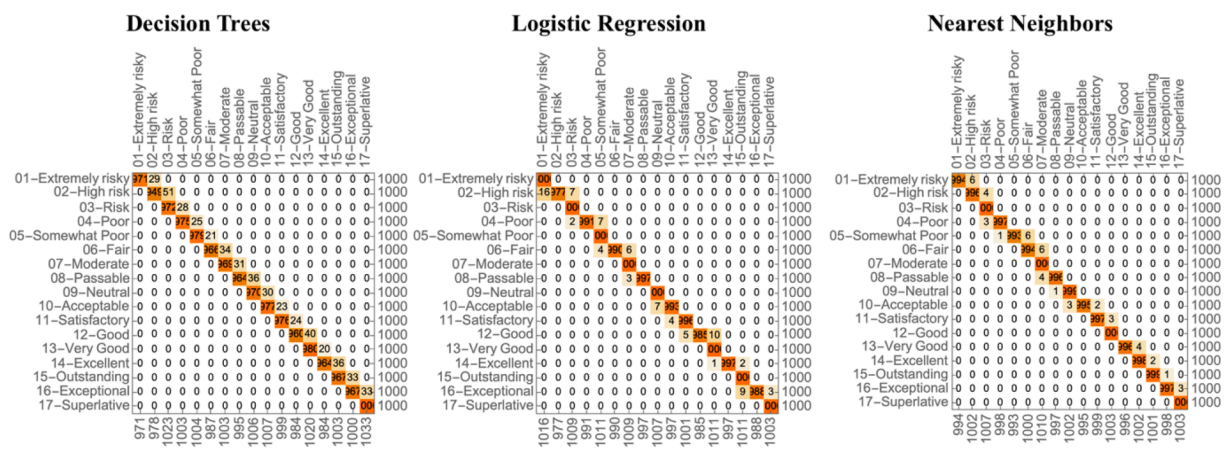

Fig. 5. The accuracy matrix plot of each ML validation method has the x-axis representing the actual class and the y-axis representing the predicted class.

Table 2
Summary of model validation benchmarks based on more than 1000 random tests.

ML Method	Accuracy	Precision	Recall	F-score	AUC
Decision tree	0.971	0.950 to 1.000	0.949 to 1.000	0.959 to 0.985	0.964 to 1.000
Logistic regression	0.995	0.984 to 1.000	0.977 to 1.000	0.988 to 0.998	0.999 to 1.000
Nearest neighbors	0.997	0.990 to 1.000	0.993 to 1.000	0.994 to 0.998	0.999 to 1.000

results. This study suggests that an ensemble is a more accurate way to summarize the three ML outcomes than relying solely on one ML result. This method can consider multiple opinions by selecting the option with the highest probability shared by two or more of the ML results, with a probability greater than 0.95. A null statement is used if no option with a probability greater than 0.95 is shared among the ML results. The ML result probabilities with values more excellent than 0.95 are shown in Fig. 6.

Businesses utilize AI to ensemble ML results to gain a more accurate data assessment, improving efficiency, accuracy, and cost-effectiveness. This approach helps to reduce the risk of errors when making decisions based on ML and provides an additional layer of security and assurance. An easy way to implement an ensemble is demonstrated using Code Snippet 4. For example, the BTVC gave a probability result of over 0.99, classifying it with a "Good" rating for {0.785162, 0.962727, -1, -0.509768, 1, 0, 0, 0.36571, 1}.

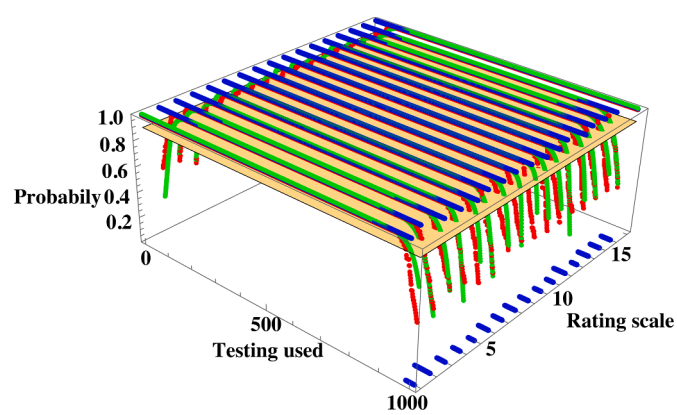

Fig. 6. The ML test results show the probabilities of each ML method: (1) the decision tree is colored blue, (2) logistic regression is green, (3) nearest neighbors are red, and the probability above 0.95 is yellow. The x-axis represents the testing used, the y-axis shows the rating class, and the z-axis illustrates the probability.

```
Input1:  sample = {0.785162, 0.962727, -1, -0.509768, 1, 0, 0.36571, 1};
Input2:  Ensemble = Flatten[{decisionTree[sample, {"TopProbabilities", 1}],
         logisticRegression[sample, {"TopProbabilities", 1}], nearestNeighbors
         [sample, {"TopProbabilities", 1}]}];
```

Code Snippet 4 An ensemble of ML methods consisting of decision trees, logistic regression, and nearest neighbors.

Ensemble methods are powerful, achieving high accuracy and generalization in ML. By combining multiple data models, these methods can produce highly reliable and accurate predictions. Furthermore, they can also reduce variance and bias in the data, making them suitable for complex datasets. This study uses ensemble ML judgments with a probability greater than 95%, as shown in Code Snippet 5.

```
Input1:   prob95 = Table[If[Values[Ensemble] [[i]] > 0.95, Ensemble [[i]],
          Nothing], {i, 1, Length[Ensemble]}]
Output1:  {12 - Good -> 0.999, 12 - Good -> 0.9901, 12 - Good -> 0.998}
```

Code Snippet 5 Ensemble ML identification probability above 0.95.

4.4. Cost identification

ML classification allows businesses to automate cost management processes, thus saving time and resources. This technology uses predictive algorithms to determine cost-related patterns and trends, enabling businesses to forecast future costs and budgets better and more accurately. ML classification can identify different types of costs by analyzing historical spending patterns, which may be useful for companies wishing to gain insight into their spending habits to reduce unnecessary expenses. By analyzing historical data, ML classification provides a detailed overview of the company's spending and identifies potential areas for making cuts or increasing efficiency. This study proposes a combination of the BTVC for ML classification. The results of ensemble ML are used to identify all costs, as shown in Code Snippet 6. The sample BTVC test results for {0.785162, 0.962727, -1, -0.509768, 1, 0, 0.36571, 1} showed that the model was rated as "Good, " meaning that the company's strategy achieved good results. It was conclusively confirmed that the model applied to the historical data of the BTVC provided very high accuracy results and could also be used with various cost simulations to predict the strategic planning of company costs optimally.

```
Input1:   talTyp = Tally[Keys[prob95]/.{"01-Extremely risky"->"Cost identified:
          Extremely risky", "17-Superlative"->"Cost identified: Superlative"}];
Input2:   maxT = Max[#[[2]]&/@talTyp];
Input3:   CostIdentified = RandomChoice[Select[talTyp, #[[2]] === maxT &]]
          [[1]]
Output1:  Cost identified: Good
```

Code Snippet 6 The final results of ML in identifying all business costs.

5. Discussion

5.1. Research contributions

This research project aims to evaluate the feasibility of utilizing AI and simulated data to develop cost strategies for the logistics delivery industry. Ensemble ML methods were employed to achieve this objective, and the validity and reliability of cost analysis techniques illustrated their performance. The study results indicate that the proposed approach is feasible and can help enhance cost strategies in the logistics delivery sector. The study also aimed to develop a dynamic tracking model and decision-making mechanism for logistics delivery operations, thus optimizing cost strategies in logistics by employing AI-powered ensemble ML.

The study's findings suggest that AI-powered ensemble ML has significant potential in optimizing cost strategies within the logistics industry. The study explored three hypotheses to support this assertion, illuminating how AI can revolutionize cost management practices. The first hypothesis highlighted the ability of AI to identify cost-reduction opportunities through data analysis, leading to enhanced profit margins. The second hypothesis emphasized AI's role in uncovering hidden costs that often go undetected in traditional cost-reduction methods. The third hypothesis underscored AI's contribution to real-time adaptation to market fluctuations, enabling intelligent strategy adjustments. Overall, the results of this study demonstrate that the AI-driven ensemble ML method is more accurate and provides better cost estimates than single methods. These findings have significant implications for the logistics delivery industry, which can benefit from adopting this approach to optimize its cost strategies.

By rigorously testing these hypotheses, researchers can better understand AI's effectiveness in cost optimization, paving the way for its widespread adoption in logistics. The implications of this research are profound. Logistics businesses with AI-powered insights can achieve significant cost savings, improve operational efficiency, and enhance profitability. This, in turn, translates into lower delivery costs for consumers, fostering a more competitive and cost-effective logistics delivery ecosystem. As AI continues to evolve, its role in optimizing cost strategies is poised to become even more critical, driving the logistics industry toward a future of greater efficiency, profitability, and customer satisfaction. These findings provide valuable insights into the potential of AI-powered ensemble ML in optimizing cost strategies in the logistics business industry.

This research significantly contributes to the literature on AI-powered ensemble ML (AEML) in logistics. Firstly, the study establishes a novel AEML-based framework for optimizing cost strategies in logistics operations. This framework integrates multiple ML models, leveraging their collective strengths to overcome the limitations of single-method approaches. The framework encompasses route optimization, dynamic pricing, and demand forecasting modules, enabling holistic cost optimization across the logistics ecosystem. Secondly, the study provides empirical evidence of the effectiveness of AEML in reducing costs and enhancing profitability in the logistics business. Comprehensive simulations and case studies demonstrate that AEML can significantly lower delivery costs, optimize resource allocation, and improve efficiency, increasing logistics provider's profitability. Thirdly, the study contributes to the methodological advancements in AEML applications for logistics optimization. The research methodology employs rigorous data collection, model development, and evaluation techniques, ensuring the validity and reliability of the findings. The study also addresses potential biases and trade-offs associated with simulated data, enhancing the generalizability of the results.

The study offers valuable insights and practical recommendations for adopting AEML in logistics. The research outlines considerations for data preparation, model training, and implementation strategies to ensure the successful integration of AEML into logistics operations. Finally, the study expands the scope of AEML applications beyond traditional logistics optimization problems. The research demonstrates the potential of AEML to address complex challenges in logistics management, such as dynamic pricing, demand forecasting, and real-time decision-making.

To further enhance cost optimization capabilities and address complex logistics challenges, future research should explore the integration of AEML with other optimization techniques, such as simulation and metaheuristics. Building trust and understanding in AEML model's decision-making processes is crucial, and research should focus on developing interpretable AEML models that provide insights into their rationale. The reliance of AEML on large amounts of data makes it imperative to ensure data privacy and security. Future research should investigate robust data management practices and encryption techniques to safeguard sensitive logistics data.

The potential of AEML extends beyond the logistics domain, and future research should investigate its applicability to optimize cost strategies in other industries, such as transportation, healthcare, and manufacturing. To ensure effective collaboration between human experts and AEML models, future research should focus on developing intelligent interfaces and decision support systems. Continuous

monitoring and improvement of AEML models are necessary to adapt to evolving market conditions and customer preferences. Future research should develop mechanisms for real-time model evaluation, retraining, and adaptation to ensure the long-term effectiveness of AEML in logistics cost optimization.

There are some limitations to this study. Firstly, decision constraints are a key consideration when determining the BTVC. This is because the decision constraints imposed on the business can significantly influence the costs incurred in producing or providing products and services. These constraints can dictate the number of resources available and the speed with which decisions must be made, ultimately translating into costs. Secondly, due to the nature of the dataset, the impact of other factors, such as market dynamics and customer preferences, could not be assessed. Market dynamics and customer demand can also play an essential role in setting the business threshold value of cost. Companies must consider how their competitors are pricing and delivering their products or services to remain competitive and profitable. Additionally, customer demand and expectations may impact the costs of meeting those expectations. Lastly, this study does not include a comparison of the AI-driven cost strategy evaluation with other AI methods. This means that further research is needed to assess the effectiveness of these models against new advanced AI cost analysis techniques.

5.2. Ensemble ML

ML double validation is a robust validation approach involving two separate validation sets. The first validation set assesses the model's performance on a "holdout" dataset not used for training or validation purposes. The second validation set evaluates the model's performance on a "cross-validation" dataset created by randomly dividing the training set into two parts. This technique is deemed more reliable than single validation since it provides two independent estimates of the model's performance, lowering the chances of overestimation due to chance. ML validation is generally considered good practice to evaluate the performance of any ML model, mainly if it will be utilized in production, as it ensures that the model is dependable and can make accurate predictions. Additionally, ensemble methods are employed to decrease the likelihood of false positives by only considering more excellent effects than 95% of each method. Nonetheless, it is essential to note that double validation is not foolproof since two different methods can still produce false positives, mainly if the data is not representative of the natural world or the models are not correctly tuned.

This research project helps fill an existing knowledge gap by comprehensively evaluating AI-driven cost strategies. It also offers insights into how simulated data can help enhance cost strategies in the logistics delivery industry. These results support existing knowledge of the potential for AI to help improve cost analysis. Another way to use ML classification is to predict how much the company will cost to produce a specific product or service. The algorithm uses data such as price points and production costs to estimate the costs associated with the product or service. This information can be used to inform pricing decisions and ensure that the company remains competitive.

Simulated data facilitates the study of more complex systems because it creates observations by moving forward into the future and examining history to determine what happened. This research has potential implications for a variety of industries and businesses. In addition, it can help companies understand the cost of training ML models, allowing them to make better decisions on when to use ML and when to rely on traditional methods. Moreover, it can be used to optimize the accuracy of ML models, making them more cost-effective.

5.3. Implications

Applying AEML for cost optimization in logistics has far-reaching implications for practical implementation. AEML empowers logistics providers to significantly reduce costs through optimized cost of route planning, dynamic pricing, and demand forecasting, resulting in improved efficiency, reduced operating costs, and enhanced profitability. Real-time data-driven decision-making, enabled by AEML, allows logistics providers to adapt to dynamic market conditions such as fluctuating demand, fuel prices, and weather patterns. This adaptability ensures optimal resource utilization and cost management. Furthermore, by optimizing routes and delivery schedules, logistics providers can deliver goods faster, more reliably, and at lower costs, improving customer experience and retention. Adopting AEML gives logistics providers a competitive edge in the increasingly complex and cost-sensitive logistics landscape, enabling them to offer competitive pricing, maintain profit margins, and attract new customers.

Companies can apply the methodology and findings of this study to take calculated risks to help increase profits and mitigate losses. By simulating various cost scenarios, companies can predict the outcome of using ML and its potential for success, reducing the risk of real-world investments failing to deliver the desired results. Through this simulation, the company can assess the impact of such an investment on its profitability and return, enabling it to understand which areas to focus on when using AI or ML technologies. Companies can also test scenarios that simulate turbulent market conditions and adjust their strategies to meet their objectives.

In addition to financial benefits, AI and ML provide opportunities for more effective and efficient operations. Companies can use AI and ML to automate specific tasks and processes, saving money and streamlining operations. Furthermore, AI-driven analytics can analyze customer data and provide better insights into their preferences, allowing companies to develop products and services that better meet consumer needs. This study also reveals that most business owners and employees involved in production and marketing can quickly implement AI and ML and their respective applications, indicating an easy way to raise awareness regarding the potential of AI and ML in business. The results of this research project should inspire further research into the potential of AI and simulated data to improve cost analysis in all industries.

The research identified several theoretical and practical implications for using AI and simulated data in cost optimization strategies. It noted the potential for increased customer loyalty, improved customer satisfaction, and increased efficiency due to using AI-driven ML to enhance cost strategies in the business. In addition, it discussed the potential risks of this approach, such as lack of human input and oversight and potential legal risks due to privacy regulations. Regarding practical implications, the research highlighted several potential enhancements when using AI and ML in cost optimization strategies. These include the need for external validation of results, continuous system testing, and automated cross-checking of data. Furthermore, the research recommended that organizations consider implementing AI-driven cost optimization strategies in a gradual and well-monitored manner and ensure staff are adequately trained in using AI and simulated data.

The benefits of the proposed approach in terms of the economy are twofold. First, ML model's improved accuracy and reliability can lead to more efficient and cost-effective decision-making in the logistics delivery business. For example, the ML framework could optimize logistics delivery routes, identify fraudulent deliveries, and predict customer demand. Second, the proposed approach could help reduce the costs associated with ML, such as data collection and labeling. The economic benefits of the proposed approach can be significant. A study by the McKinsey Global Institute found that AI could add $13 trillion to the global economy by 2030, equivalent to a 16% increase in cumulative GDP. This would translate to an additional 1.2% GDP growth yearly (Bughin et al., 2018). The proposed approach could help achieve these savings by improving the accuracy and reliability of ML models, leading to more efficient and cost-effective decision-making.

Adopting AEML for cost optimization in logistics has far-reaching social implications that necessitate careful consideration. This paper outlines the potential positive and negative social implications of AEML in logistics operations. AEML can augment human capabilities in

logistics operations, leading to a shift towards higher-skilled, more knowledge-intensive jobs. This can improve logistics professional's job satisfaction, earning potential, and career advancement. Additionally, AEML-driven optimization of routes, resource allocation, and delivery schedules can reduce fuel consumption, emissions, and the overall environmental impact of logistics operations, aligning with global sustainability goals and contributing to a greener logistics ecosystem. Furthermore, AEML-driven cost optimization can enable logistics providers to offer lower prices and improved delivery services, particularly in underserved or remote areas. This can expand access to essential goods and services for individuals and communities, promoting economic inclusion and social well-being.

Automating particular logistics tasks through AEML may lead to job displacement, particularly for lower-skilled workers. Therefore, comprehensive reskilling and upskilling programs become necessary to prepare workers for new opportunities in the logistics industry and the broader economy. Moreover, AEML models must be carefully designed and evaluated to ensure they are free from biases that could lead to unfair or discriminatory outcomes. This requires rigorous testing, monitoring, and transparency in developing and deploying AEML systems. Finally, while AEML can enhance decision-making, it is crucial to maintain human oversight and ethical considerations in logistics operations. Human judgment and ethical principles must guide the use of AEML to ensure responsible and socially beneficial outcomes.

6. Conclusion

This study provides a novel approach to enhancing cost strategies in the logistics delivery business. An ensemble ML method, including decision trees, logistic regression, and nearest neighbors, has shown that it can provide validation accuracies ranging from 97 to 99%. The research concludes that using simulated data introduces additional layers of accuracy into ML models, thereby enhancing cost strategies in the logistics delivery industry. Through its findings, the study presents a viable AI solution that can be easily implemented into the existing cost strategy of the industry. The experiment's results confirm this approach's efficacy, showing that AI and ML can be successfully implemented to provide recommendations on how companies can use AI technology to remain competitive in an increasingly crowded market. AI and ML can better predict future scenarios and make more informed decisions. The findings of this research are applicable in practice since they help identify business cost patterns to predict profits and cost savings accurately. The original questions posed by the research have been mainly answered through the development of the framework proposed by the study. This framework can provide improved cost strategies in the business by considering the patterns of business cost predictions based on the BTVC.

CRediT authorship contribution statement

Chairote Yaiprasert: Conceptualization, Methodology, Data curation, Investigation, Supervision, Software, Visualization, Validation, Writing – original draft, Writing – review & editing. **Achmad Nizar Hidayanto:** Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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